1 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/scm_verify/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5` `ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`
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1.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/scm_verify`
- Number of saved time samples: 13
- Number of profile snapshots: 1
- Solver accepted/rejected steps: 61 / 3
- RHS evaluations: 8017
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

1.2 Numerical Verification

- Maximum surface energy closure error: $0\,\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.719025, 108.431948]\ \mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.035577, 29.766726]$

1.3 Generated Artifacts

- Time-series CSV: `results/scm_verify/time_series.csv`
- Diagnostic summary JSON: `results/scm_verify/summary.json`
- Payload JLD2: `results/scm_verify/payload.jld2`

1.4 Figure Manifest

1. Time series of T__s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times
5. Surface energy budget components (R__n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e__xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

1.5 Included Plots

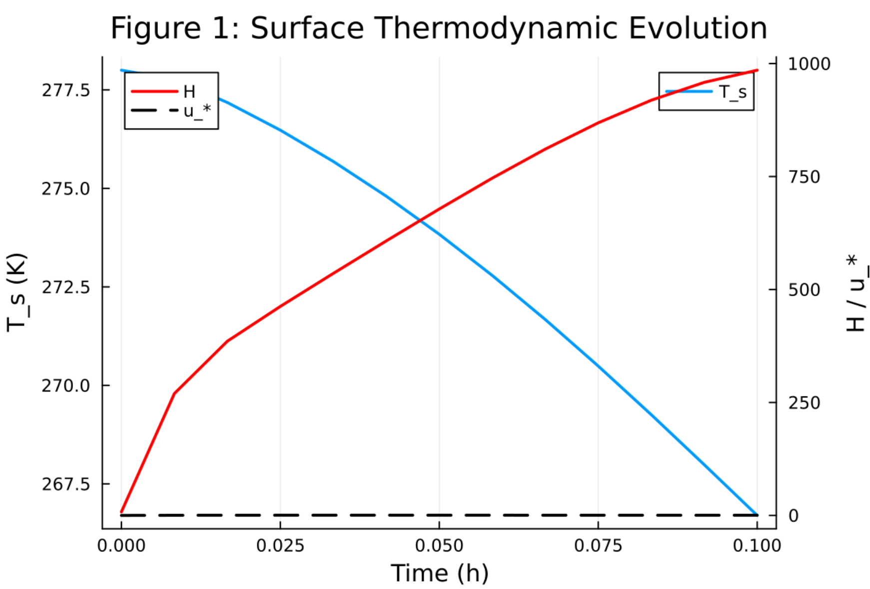

Figure 1: Time series of T__s, sensible heat flux, and friction velocity

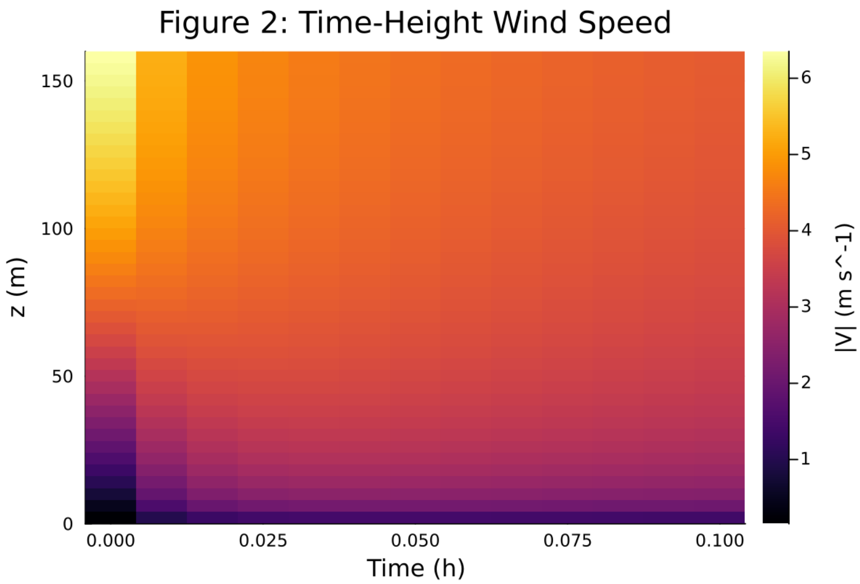

Figure 2: Time-height contour of wind speed with LLJ evolution

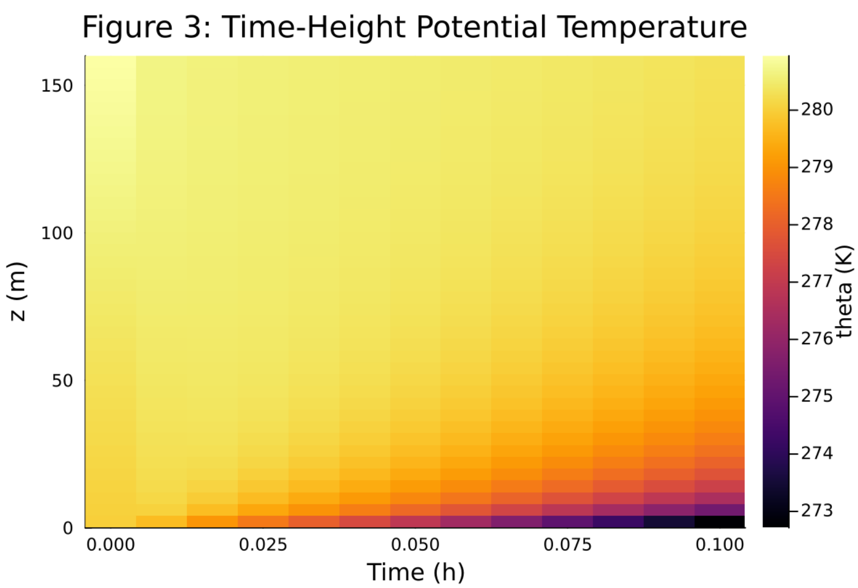

Figure 3: Time-height contour of potential temperature and inversion growth

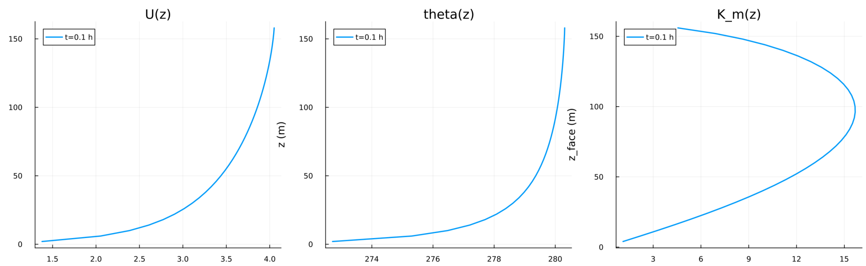

Figure 4: Vertical profiles of U, theta, and Km at representative times

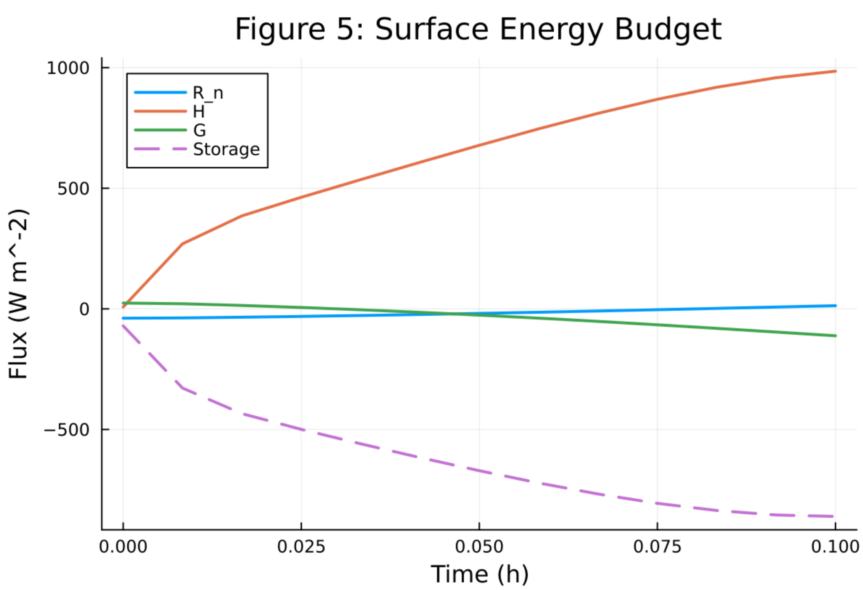

Figure 5: Surface energy budget components (R__n, H, G, storage)

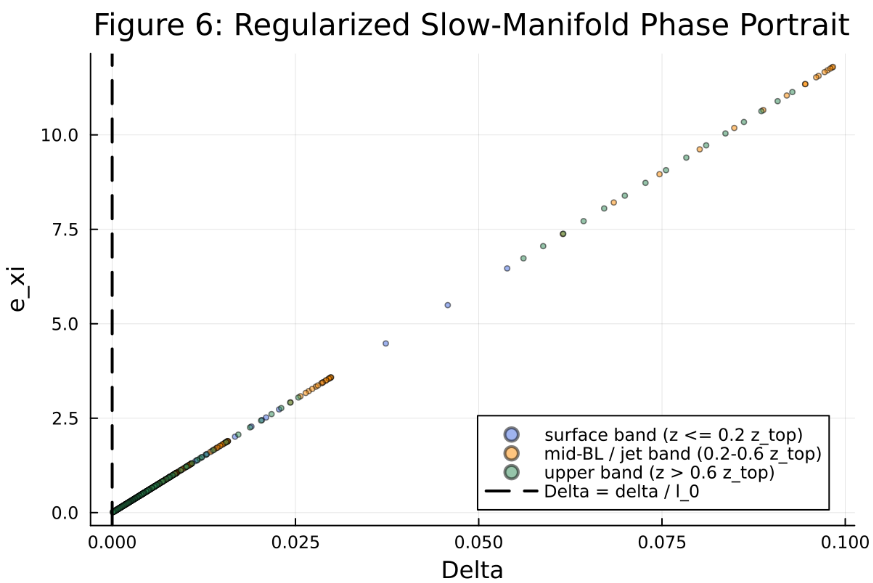

Figure 6: Phase portrait on regularized manifold (Delta vs e__xi)

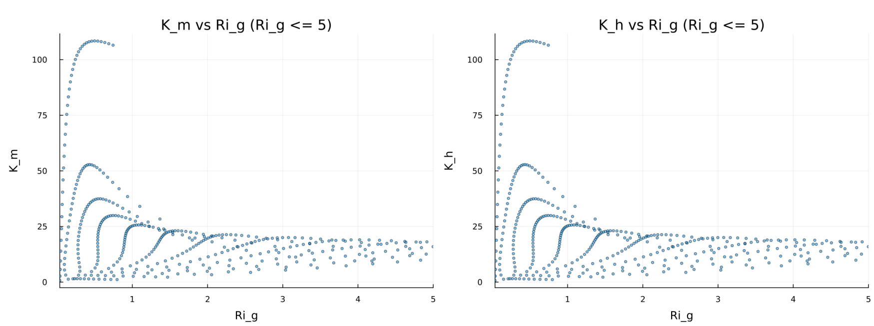

Figure 7: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

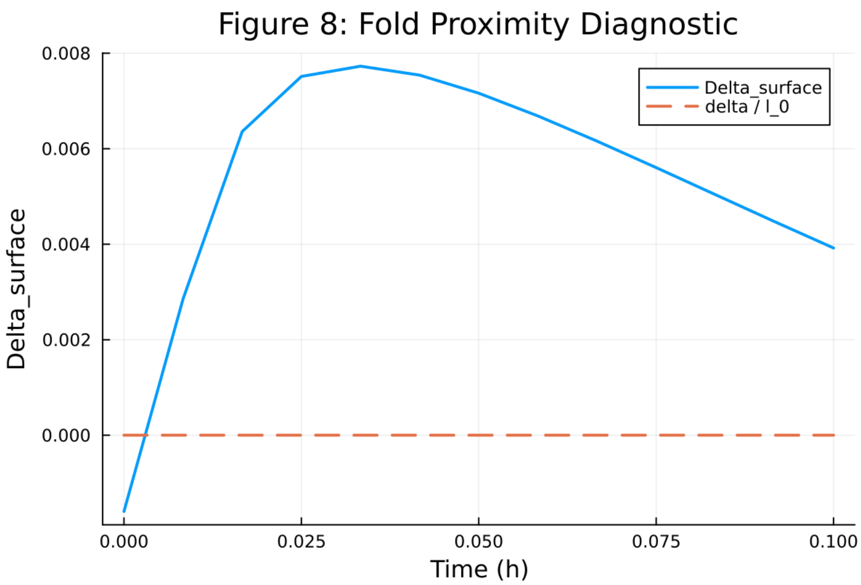

Figure 8: Fold proximity diagnostics versus time